



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO L.K. FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

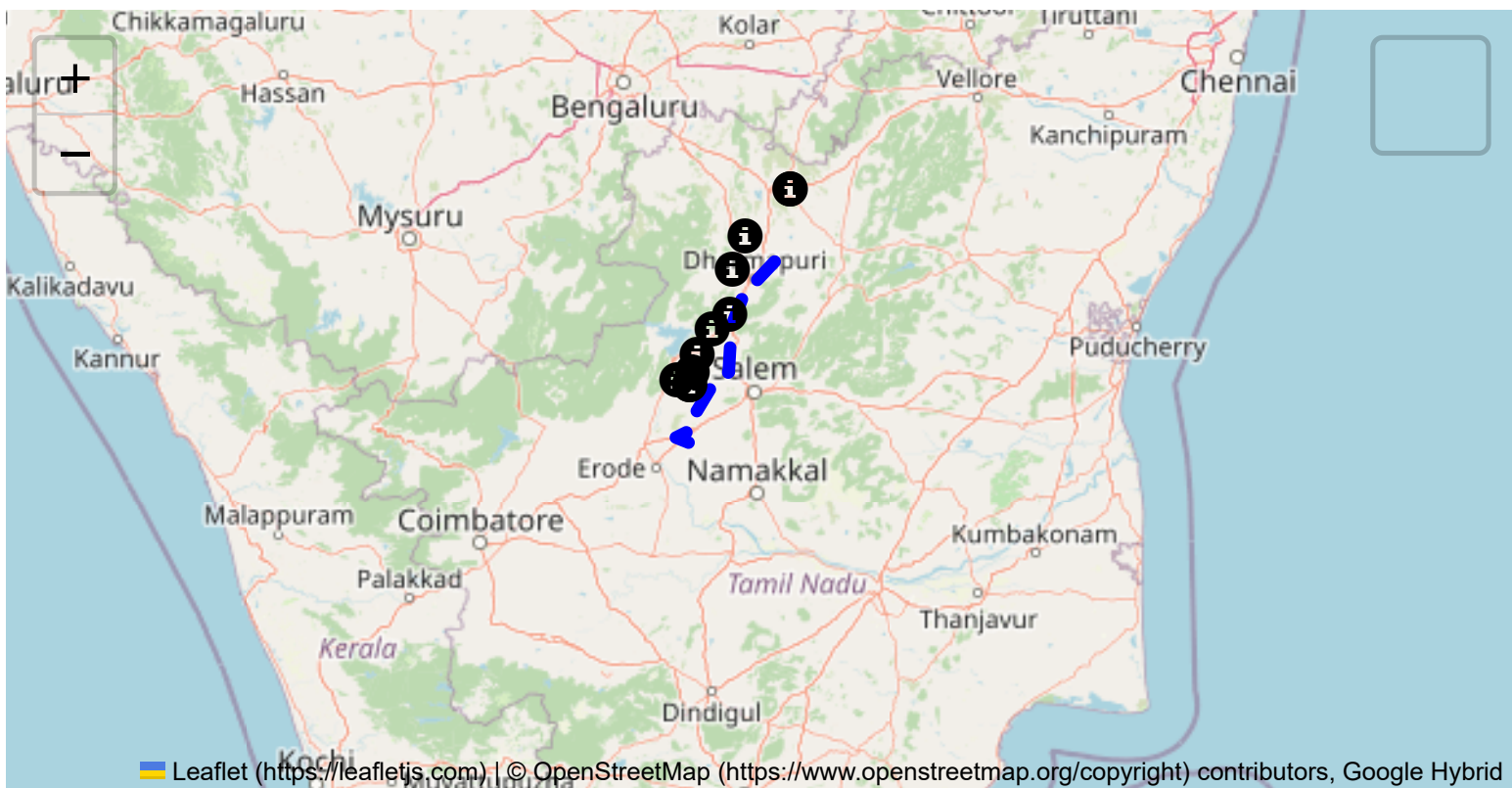
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

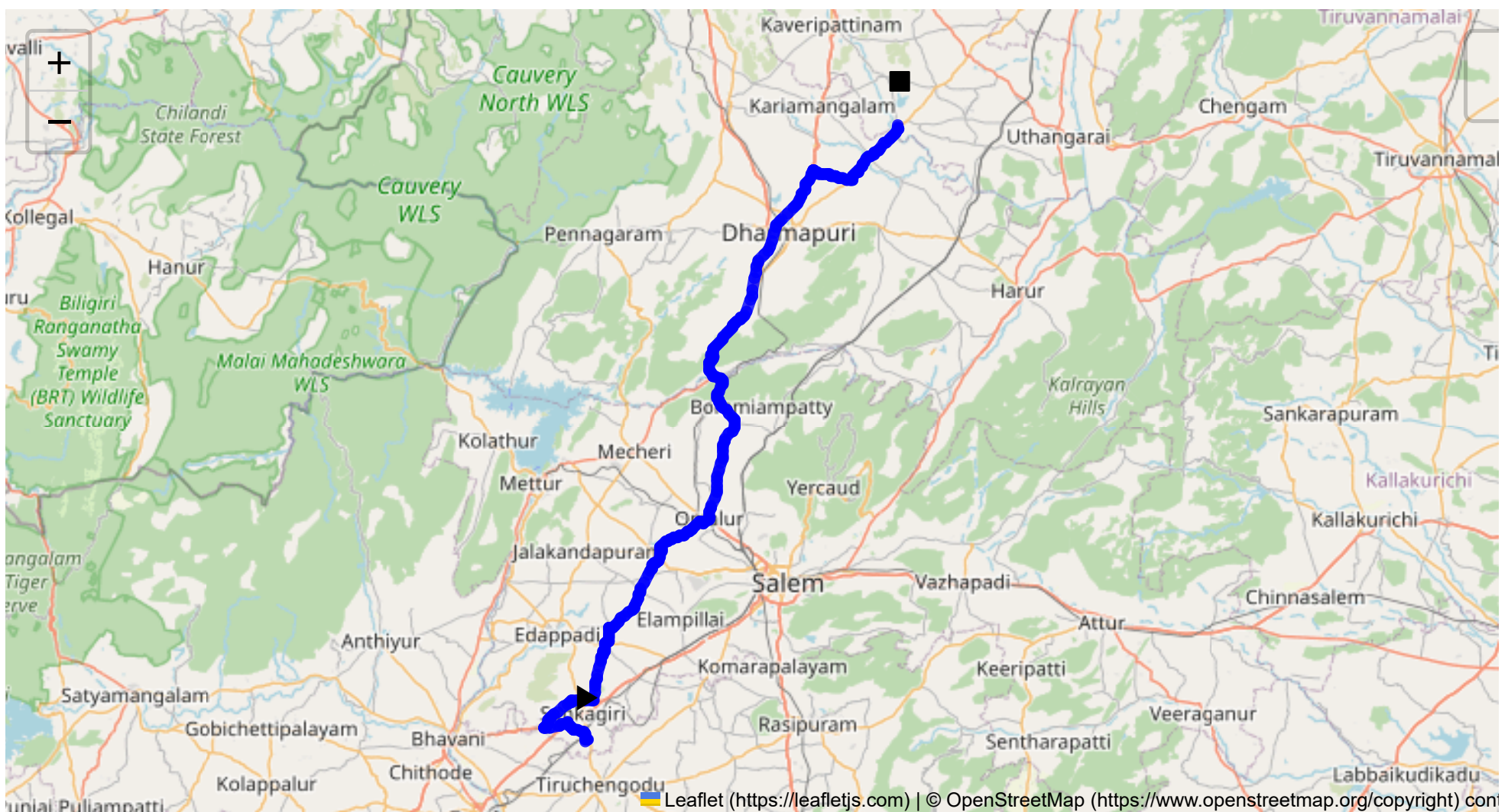
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 133.11 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.280423, 78.310049)



Welcome to the Journey Risk Management Study

Below is a detailed safety analysis of the route from CVQF+23W, Sangagiri to 78J5+CRR, Barur in Tamil Nadu, covering approximately 133.11 kilometers:

1. Overview of the Route Map:

- The route traverses through various small towns and connects major highways, such as the Salem - Kochi Highway and SH 86. The journey involves transitioning from local roads to more

heavily trafficked highways, presenting a mix of rural and semi-urban driving conditions.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**

- **Weather:** Tamil Nadu typically experiences a tropical climate. The route may be affected by heavy monsoon rains from October to December, potentially leading to flooding and reduced road visibility.
- **Hazards:** During the summer months, extreme heat can affect engine performance and tire pressure.

3. **Analysis of Traffic Patterns:**

- **Peak Hours:** Morning (7 AM - 9 AM) and evening (5 PM - 7 PM) commuters can cause congestion, particularly in Omalur and Dharmapuri regions.
- **Congestion-Prone Areas:** Notable traffic bottlenecks can occur near the junctions of Bhavani Main Road and Salem - Kochi Highway.

4. **Assessment of Road Quality and Infrastructure:**

- Some local roads, especially near smaller towns, may be narrow and poorly maintained. The main highways are generally well-paved but may have deterioration due to heavy rains.

5. **Suggestions for Alternative Routes for Emergencies:**

- Consider using the NH 44 (part of the North-South Corridor) for quicker access in emergencies, avoiding smaller roads close to smaller towns.

6. **Summary of Local Regulations Affecting Hazardous Material Transport:**

- Tamil Nadu has regulations stipulating specific time windows for hazardous materials transport, typically avoiding peak city traffic hours. Compliance with the Motor Vehicles Act for hazardous material signage is mandatory.

7. **Overview of Historical Incidents:**

- There have been incidents involving overturned trucks due to sharp turns and speeding. Road awareness programs are periodically conducted due to the presence of hazardous industries in the region.

8. **Environmental Considerations and Sensitive Areas:**

- The route avoids major wildlife sanctuaries, but it is essential to minimize emissions and spillage risks to prevent environmental impact.

9. **Analysis of Communication Coverage:**

- Major stretches, especially on highways and through towns, generally have reliable mobile network coverage. Some rural sections may experience brief dead zones.

10. **Estimated Emergency Response Times:**

- **Urban Areas (e.g., Thoppur, Dharmapuri):** Approximately 20-30 minutes.
- **Rural Areas:** Emergency response may take up to 45 minutes to an hour due to distance and road conditions.

11. **Overall Summary of Risk Assessment:**

- The route features a combination of well-maintained highways and less reliable rural roads. Weather conditions, especially during the monsoon, pose a significant risk. Traffic congestion in semi-urban areas requires careful planning. Adherence to local regulations and an awareness of environmental and safety practices are critical for mitigating risks associated with the transport of hazardous materials.

In summary, this route requires a strategic approach to minimize risks, particularly considering weather patterns, road conditions, and local traffic regulations. Frequent updates from local authorities on road conditions and incidents are recommended to ensure a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45543, 77.81740	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	High	11.70233, 77.97869	15 KM/Hr
19	Turn	Medium	11.71394, 77.99572	30 KM/Hr
20	Turn	Medium	11.71916, 78.01117	30 KM/Hr
21	Turn	Medium	11.73787, 78.03528	30 KM/Hr
22	Turn	High	11.73651, 78.03868	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
23	Turn	Medium	11.74257, 78.04708	30 KM/Hr
24	Turn	Medium	11.86805, 78.08353	30 KM/Hr
25	Turn	Medium	11.87913, 78.08206	30 KM/Hr
26	Turn	Medium	11.92566, 78.06503	30 KM/Hr
27	Turn	Medium	11.93694, 78.05425	30 KM/Hr
28	Turn	Medium	11.95531, 78.04698	30 KM/Hr
29	Turn	High	11.95803, 78.04964	15 KM/Hr
30	Turn	Medium	11.96068, 78.04960	30 KM/Hr
31	Turn	Medium	11.96370, 78.05142	30 KM/Hr
32	Turn	Medium	11.96433, 78.05113	30 KM/Hr
33	Turn	High	11.96542, 78.04938	15 KM/Hr
34	Turn	High	11.96650, 78.04927	15 KM/Hr
35	Turn	High	12.21894, 78.19193	15 KM/Hr
36	Turn	Medium	12.21623, 78.19627	30 KM/Hr
37	Turn	High	12.21372, 78.21163	15 KM/Hr
38	Turn	High	12.21458, 78.21217	15 KM/Hr
39	Turn	High	12.21205, 78.21630	15 KM/Hr
40	Turn	Medium	12.21231, 78.21741	30 KM/Hr
41	Turn	Medium	12.21111, 78.22099	30 KM/Hr
42	Turn	Medium	12.21146, 78.22233	30 KM/Hr
43	Turn	High	12.21132, 78.22306	15 KM/Hr
44	Turn	Medium	12.21037, 78.22363	30 KM/Hr
45	Turn	Medium	12.20519, 78.24472	30 KM/Hr
46	Turn	Medium	12.20811, 78.24769	30 KM/Hr
47	Turn	Medium	12.24528, 78.28177	30 KM/Hr
48	Turn	Medium	12.26516, 78.30306	30 KM/Hr
49	Turn	Medium	12.26736, 78.30396	30 KM/Hr
50	Turn	Medium	12.26988, 78.30727	30 KM/Hr
51	Turn	Medium	12.27533, 78.31023	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
28	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56083, 77.89921



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56180, 77.90019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59328, 77.90809



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70233, 77.97869



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71394, 77.99572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71916, 78.01117



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73787, 78.03528



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03868



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708

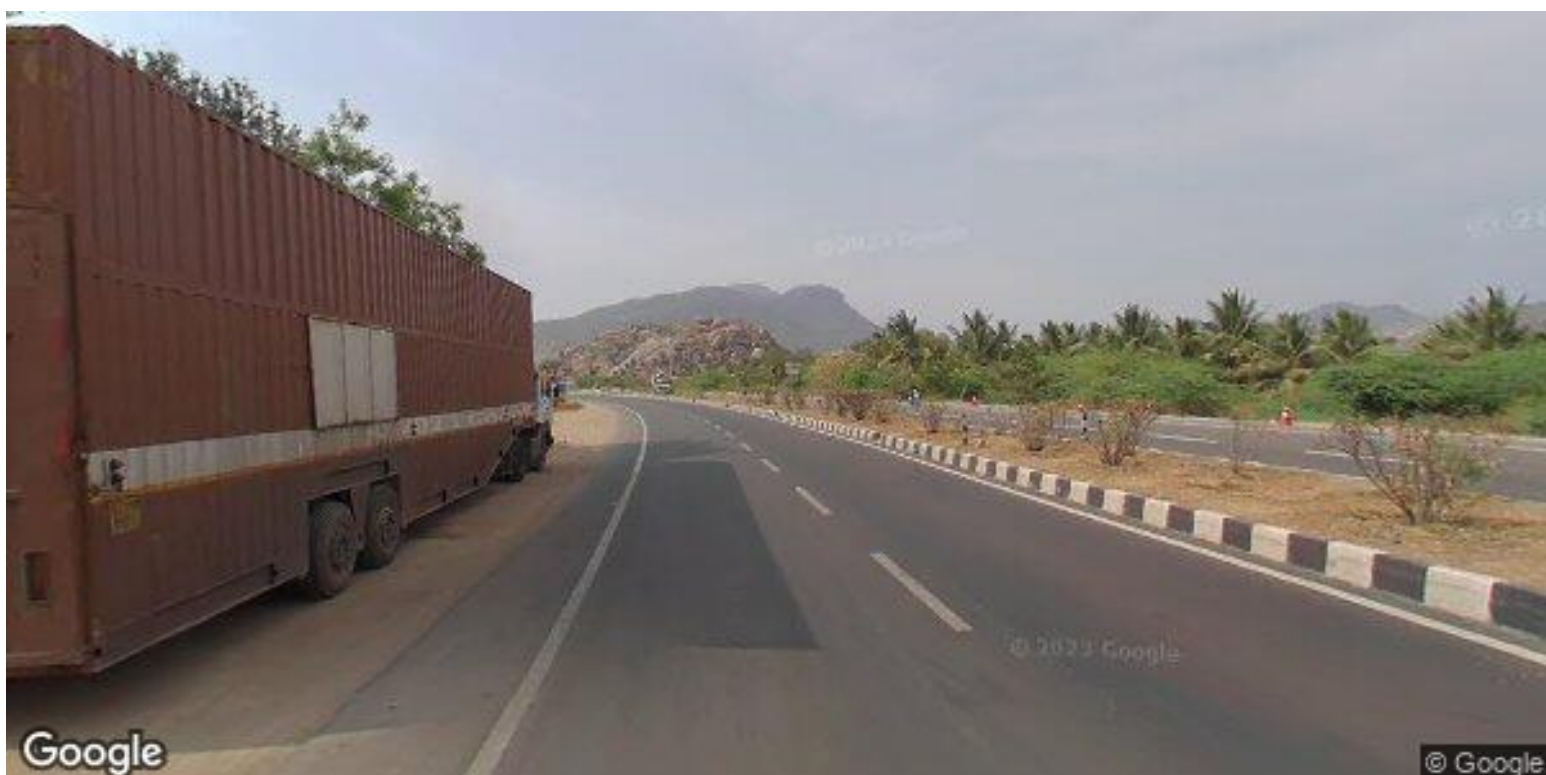


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86805, 78.08353



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93694, 78.05425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95531, 78.04698



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95803, 78.04964



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96068, 78.04960



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21894, 78.19193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.21623, 78.19627



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21372, 78.21163



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21458, 78.21217



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.21205, 78.21630



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21146, 78.22233



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.21132, 78.22306



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21037, 78.22363



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.20519, 78.24472



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.20811, 78.24769



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24528, 78.28177



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26516, 78.30306



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26736, 78.30396



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.26988, 78.30727



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.27533, 78.31023
